8 **(a)** Force on sail $= (\Delta v)\left(\frac{Nm}{t}\right)$ or $= \left(\frac{N}{t}\right)\Delta p$ M1

$\Delta p \left(= 2mc = 2\frac{h}{\lambda}\right) = 2 \times 6.63 \times 10^{-34} / 1.0 \times 10^{-6}$ C1

$\frac{N}{t} = \frac{P}{hf} = \frac{P}{hc} \times \lambda = (100 \times 10^9 \times 1.0 \times 10^{-6}) / (6.63 \times 10^{-34} \times 3 \times 10^8)$ C1

$F = (2 \times 6.63 \times 10^{-34} / 1.0 \times 10^{-6}) \times (100 \times 10^9 \times 1.0 \times 10^{-6}) / (6.63 \times 10^{-34} \times 3 \times 10^8)$

$F = 666.7 = 670$ N A0

Award one mark if candidate uses $F = 2P / c$.

(b) **(i)** acceleration $= 670 / 4.0 \times 10^{-3} = 1.7 \times 10^5$ m s^{-2} (accept 1.67 to 1.7) A1

(ii) $t = (v - u) / a = 0.20 \times 3 \times 10^8 / 1.675 \times 10^5 = 358$ s (358 – 360 s) A1

(c) **(i)** $\theta = s / r = 5.0 / 1.0 \times 10^{10}$ M1

$\theta = 5.0 \times 10^{-10}$ rad A0

(ii) laser that is not incident on the sail is wasted power B1

(iii) either $\sin (5.0 \times 10^{-10} / 2) = \lambda / b \Rightarrow b = 1.0 \times 10^{-6} / \sin (5.0 \times 10^{-10} / 2) = 4000$ m
or (for small angle) $b = \lambda / \frac{1}{2}\theta = 1.0 \times 10^{-6} / 2.5 \times 10^{-10} = 4000$ m A1

(d) power absorbed $= 0.001 \times 100 \times 10^9 = 100 \times 10^6$ W M1

thermal equilibrium: power absorbed = power radiated $= \varepsilon\sigma AT^4 = 100 \times 10^6$ M1

$T = [100 \times 10^6 / (0.80 \times 5.67 \times 10^{-8} \times 2 \times \pi \times 2.5^2)]^{1/4} = 2740$ K or 2700 K A1

(e) low particle density / vacuum in space, hence sail cannot lose heat through conduction and convection B1

(f) **(i)** Only a fraction (or part) of the sail's surface is illuminated (by the laser) B1

(ii) Adjust beam width of the laser by shutting down some of the laser emitters B1

reduce the effective size of the array (hence changing the angle of diffraction of the beam or beamwidth) B1

(g) useful energy per decay $= 0.067 \times 5.48 \times 10^6 \times 1.60 \times 10^{-19} = 5.875 \times 10^{-14}$ J C1

activity needed after 30 years $A = 5.0 \times 10^{-3} / 5.875 \times 10^{-14} = 8.511 \times 10^{10}$ s^{-1}

number of nuclei after 30 years $N = A / \lambda = 8.511 \times 10^{10} / (\ln 2 / 2.72 \times 10^9) = 3.340 \times 10^{20}$ C1

number of nuclei initially $N_0 = N / e^{-\lambda t} = 3.340 \times 10^{20} / \exp(-\ln 2 \times 30 / 86.4) = 4.249 \times 10^{20}$ C1

initial mass of plutonium = $N_0 \times$ mass of nuclei $= 4.249 \times 10^{20} \times 238 \times 1.66 \times 10^{-27}$ M1

$= 1.7 \times 10^{-4}$ kg = 170 or 168 mg A1